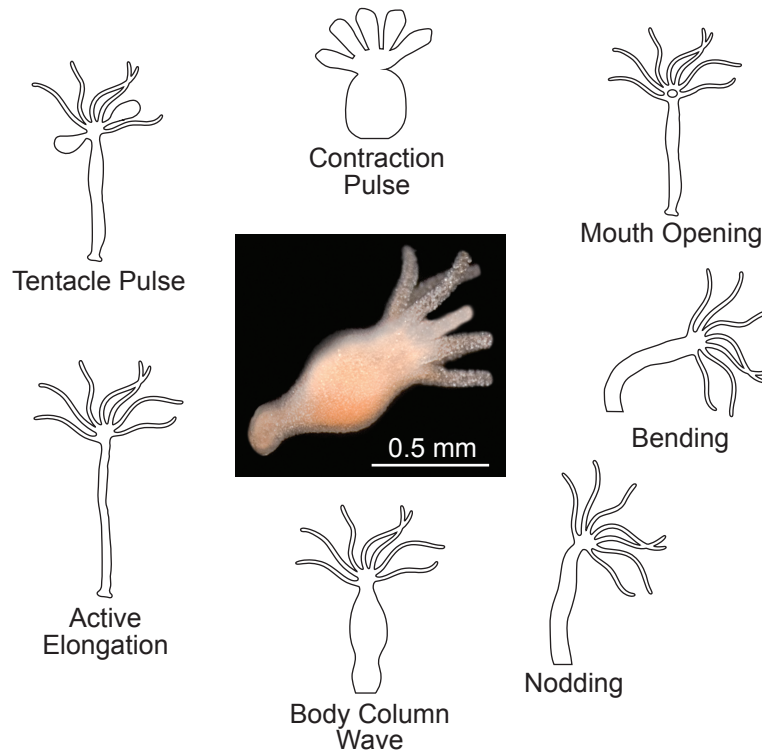


4. **Image 4.1** shows a photograph of a live *Hydra* and drawings of a range of activities that may be observed in *Hydra*. The activities shown take between 3 seconds and 17 seconds to complete.

Image 4.1



- (a) Name the type of nervous system in *Hydra* and explain why its movements take a relatively long time. [2]

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In humans, nerve cells are differentiated to carry out specific functions.

Image 4.2 is a diagram of a specialised nerve cell. The rectangle represents the area that is shown in **Image 4.3**.

Image 4.3 is an electron micrograph of the tip of a specialised nerve cell together with part of an adjacent cell.

Image 4.2

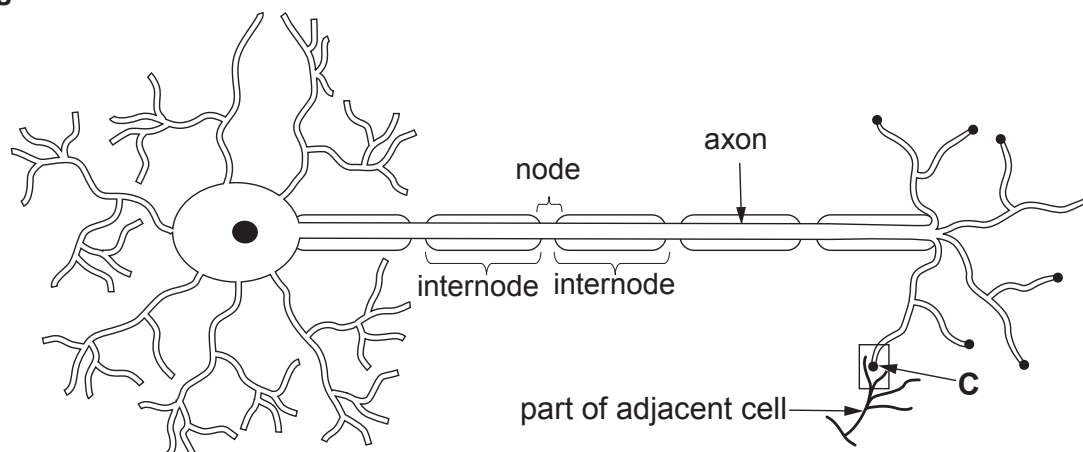
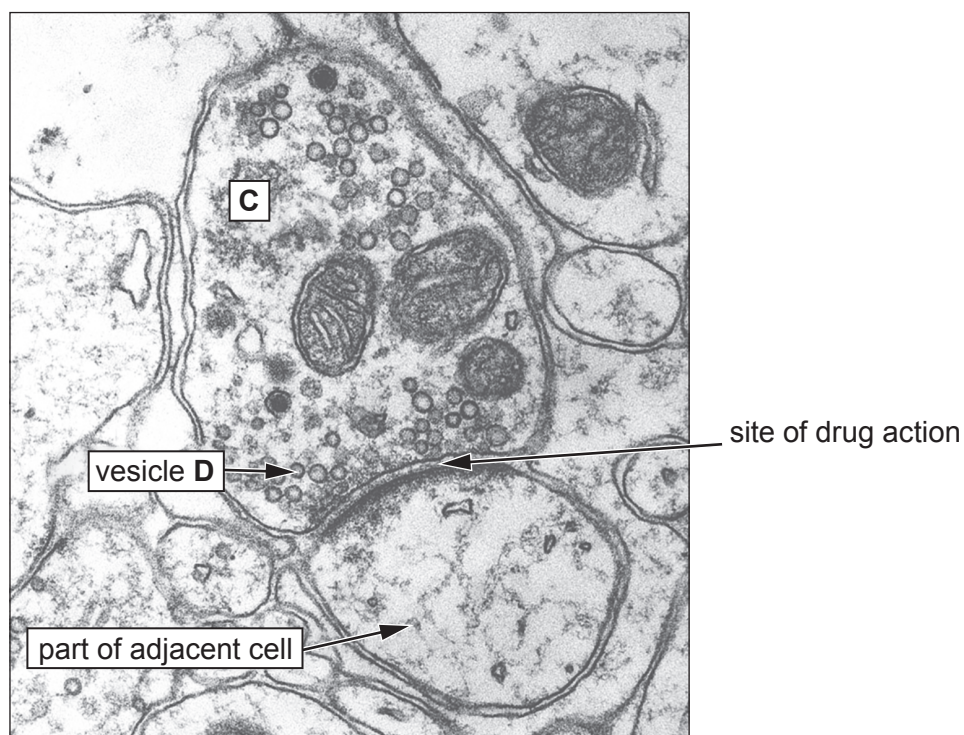


Image 4.3



- (b) (i) Name the structure labelled **C** in **Images 4.2** and **4.3**. [1]

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- (ii) Name the type of vesicle labelled **D** in **Image 4.3** and describe its function. [1]

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- (iii) Some drugs are described as acetylcholinesterase inhibitors. The site of action of these drugs is labelled on **Image 4.3**.
Suggest why these drugs cause over-stimulation of the membrane of the adjacent cell. [2]

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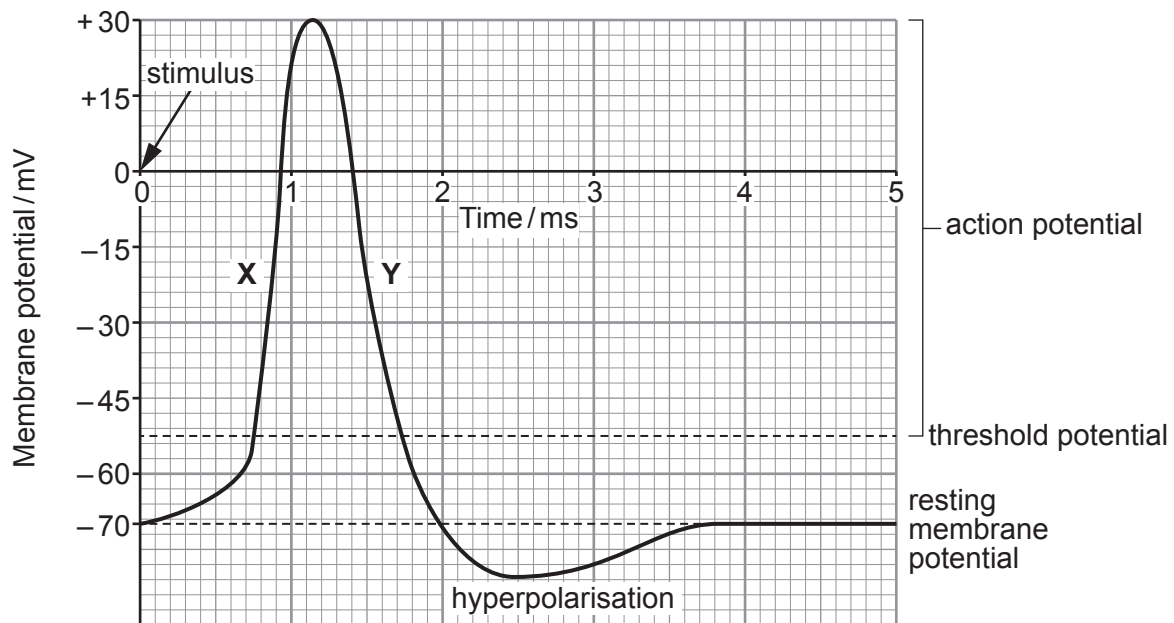
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Graph 4.4 shows an oscilloscope trace from an experiment to measure the potential difference across the membrane of a squid giant axon.

Graph 4.4



- (c) (i) Name the processes represented by letters **X** and **Y** in **Graph 4.4**. [1]

X **Y**

- (ii) With reference to **Graph 4.4**, explain how voltage-gated sodium ion channels bring about process **X**. [2]

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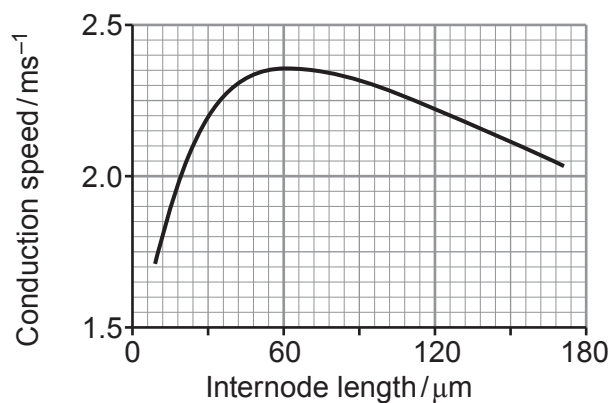
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The speed at which nerve impulses are conducted along nerve fibres depends on a number of factors, including myelination. **Graph 4.5** shows the results of a computer simulation to investigate the effect of length of myelin internode (see **Image 4.2**) on conduction speed.

Graph 4.5



- (d) Describe the relationship between internode length and conduction speed and use your knowledge of nerve impulse transmission to explain the results. [4]

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